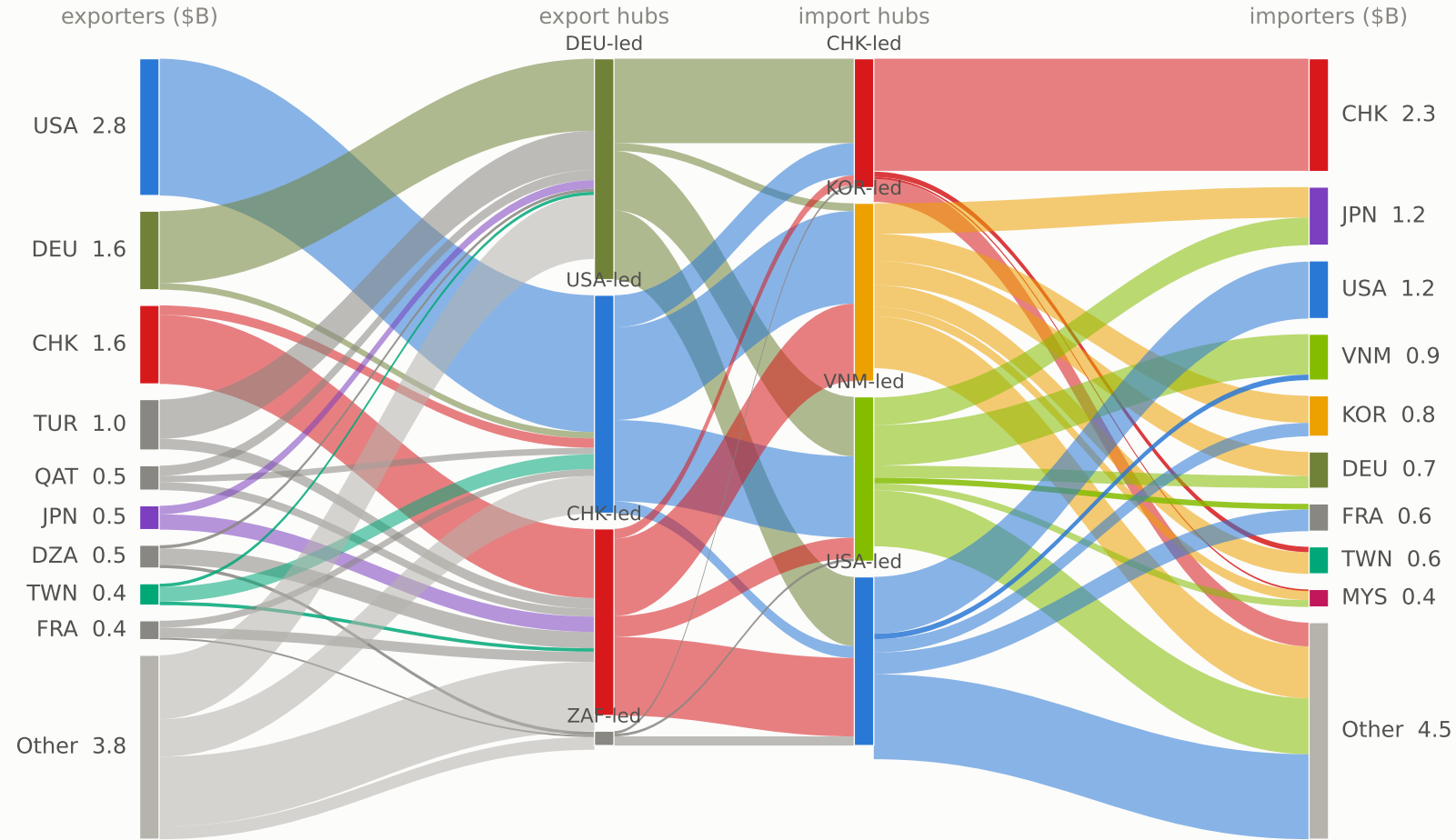


Stage 1 — Raw materials (silicon 280461, rare gases, Ga/Ge, chemicals) — 2024
 through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$13B



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 on Atlas bilateral data); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral, all countries.